

DEVELOPMENT OF DIGITAL INVENTORY
SYSTEM FOR KITCHEN ITEM AT SERAI
PARADIGM OUTLET

IZZAT ISKANDAR SHAH BIN SAHARIJAT

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

DEVELOPMENT OF DIGITAL INVENTORY SYSTEM FOR KITCHEN ITEM AT
SERAI PARADIGM OUTLET

IZZAT ISKANDAR SHAH BIN SAHARIJAT

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Universiti Tun Hussein Onn Malaysia

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I hereby declare that the work in this project report is my own except for quotations
and summaries which have been duly acknowledged

Student :
IZZAT ISKANDAR SHAH BIN SAHARIJAT
Date :

Supervisor :
LISA MARIAM SYAWAL BINTI ISMAIL

Supervisor :
CO-Supervisor Name

DEDICATION

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ABSTRACT

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CHAPTER 1

INTRODUCTION

1.1 Research background

The world has undergone multiple industrial revolutions that have reshaped industries, including hospitality and food & beverage (F&B). These transformations evolved from the First Industrial Revolution (IR1.0), which introduced mechanization, to the Second (IR2.0), which emphasized mass production, followed by the Third (IR3.0) with the use of electronics and automation, and now the Fourth Industrial Revolution (IR4.0). Each phase signifies progress toward greater efficiency, productivity, and service innovation—key elements in the competitive hospitality and F&B sectors.

Today, the hospitality and F&B industry is experiencing the rapid changes of IR4.0, an era marked by smart technologies that transform customer experiences and business operations. Artificial Intelligence (AI) and Machine Learning (ML) are increasingly applied in customer service, such as chatbots for reservations, personalized menu suggestions, and predictive demand forecasting to reduce food waste. The Internet of Things (IoT) allows for smart kitchens and connected appliances that monitor cooking, storage, and inventory in real time, ensuring consistency and safety in food quality. Cloud-based systems and Big Data provide valuable insights into consumer preferences, peak dining hours, and supply chain management, helping businesses make data-driven decisions that enhance efficiency.

In addition, IR4.0 technologies enable greater flexibility and sustainability within the industry. Digital payment systems, mobile ordering, and self-service kiosks improve convenience, while energy-efficient smart appliances and digital monitoring systems reduce environmental impact. This not only improves productivity but also aligns with the growing demand for sustainable dining and hospitality practices.

Therefore, just as past industrial revolutions transformed production and service models, the hospitality and F&B industry must continue to adapt to the IR4.0 landscape. By embracing technological advancements, businesses can remain competitive, deliver superior guest experiences, reduce operational challenges, and sustain economic growth in a fast-evolving global market.

1.2 Problem statement

Despite the rapid development of Industry 4.0, numerous small, medium and big scale enterprises within the hospitality and food & beverage (F&B) sector have yet to align themselves with the transformations brought about by the industrial revolution. This situation arises primarily because the businesses often struggle to implement advanced technologies due to high investment costs, lack of technical expertise, and the overwhelming pace of technological evolution. As a result, many firms, particularly those operating small, medium or big scale such as local restaurants, cafés, and catering services, remain entrenched in outdated operational paradigms.

In the context of hospitality and F&B, the inability to adopt digital transformation is especially visible in the area of inventory management. Inventory plays a pivotal role in this sector because the success of food operations depends heavily on maintaining sufficient and accurate stock levels of raw ingredients, beverages, and consumables. Without effective inventory control, businesses risk running out of essential supplies during peak service hours, overstocking items that may spoil, or mismanaging costs, all of which negatively affect both customer satisfaction and financial performance.

Currently, many businesses still rely on manual systems of inventory computation, which often involve handwritten records, spreadsheets, or basic tallying methods. While these traditional approaches have been in practice for decades, they pose significant challenges in the modern industry environment. Manual processes are prone to human errors, such as miscounting, misreporting, and data entry mistakes, which can lead to incorrect stock levels. Additionally, manual systems lack real-time visibility and integration with other operational processes, making it difficult for managers to anticipate shortages, prevent wastage, or track usage patterns accurately.

In the highly competitive hospitality industry, where efficiency and accuracy are key to sustaining customer loyalty, these shortcomings can create long-term operational inefficiencies.

Inventory mismanagement in the F&B industry can also have a direct impact on customer service quality. For example, a restaurant that frequently runs out of popular menu items due to poor stock monitoring may frustrate guests and harm its reputation. Similarly, over purchasing raw ingredients without a systematic method of tracking demand may lead to unnecessary food waste, increased expenses, and ultimately reduced profit margins. This demonstrates the critical importance of inventory management not only as an internal control system but also as a customer-facing element of service delivery.

To address these issues, researchers have increasingly highlighted the need for a digital inventory system that replaces manual stock counts. A digital system can automate the tracking of supplies, monitor real-time stock levels, and generate accurate records accessible to managers at any time. By utilizing barcode scanning, smart sensors, and cloud-based platforms, digital inventory solutions can systematically minimize potential errors and ensure methodical management of stock. This aligns with the broader objectives of Industry 4.0, which emphasizes automation, data-driven decision-making, and technological integration across all levels of business operations.

The implementation of a digital inventory system offers multiple benefits to the hospitality and F&B industry. Firstly, it enhances efficiency by reducing the time staff spend on manual counting and reporting, thereby allowing employees to focus more on service quality and guest satisfaction. Secondly, it provides data accuracy that supports better decision making in procurement, menu planning, and cost control. For example, data analytics embedded in digital systems can identify peak demand periods, predict ingredient usage, and suggest optimal reorder levels. This reduces the risk of overstocking and understocking, ensuring that operations run smoothly and consistently.

Moreover, digital inventory systems contribute to sustainability efforts within the industry. Food waste is a persistent issue in F&B, and better stock monitoring through digital tools can reduce unnecessary spoilage by aligning purchasing decisions with actual demand. This not only reduces costs but also supports environmentally

responsible business practices, which are increasingly valued by modern consumers. By adopting digital inventory system, businesses can therefore strengthen their competitive edge while also aligning with global sustainability goals.

However, the transition to digital inventory system does not come without challenges. The enterprises, in particular, may face obstacles such as initial investment costs, the need for staff training, and resistance to changing long-standing practices. Yet, as Industry 4.0 technologies become more affordable and accessible, these barriers are gradually being reduced. Cloud-based inventory platforms, for example, are often available on subscription models that minimize upfront expenses, making them feasible for small restaurants and food outlets. Additionally, mobile-based applications are user-friendly and do not require extensive technical knowledge, allowing staff to adapt quickly to new systems.

In conclusion, while many businesses in the hospitality and F&B industry continue to rely on outdated manual systems, the growing demands of Industry 4.0 make digital transformation a necessity rather than an option. The development and adoption of digital inventory systems present a vital step towards improving efficiency, reducing errors, minimizing waste, and enhancing overall service quality. As the industry continues to evolve in response to technological advancements, embracing digital solutions will not only ensure operational sustainability but also strengthen competitiveness in an increasingly dynamic and challenging marketplace.

1.3 Research objective

The objective of this project is to develop and evaluate a digital inventory system that is specifically designed to address the unique challenges faced in the hospitality and food & beverage (F&B) industry. Inventory management plays a critical role in ensuring smooth kitchen operations, cost control, and service quality. Within this industry, effective inventory systems not only help document and monitor stock but also serve as essential tools for minimizing waste, improving efficiency, and maintaining consistent product quality for customers.

The specific objectives of this project reflect the need to solve these operational problems while improving long-term sustainability and efficiency in the F&B industry:

1. To develop a digital inventory system for kitchen items at Serai Paradigm outlet
2. To evaluate the effectiveness of the digital inventory system developed at Serai Paradigm outlet

Overall, the project objectives are closely tied to the practical needs of the hospitality and F&B industry, where efficient inventory management directly influences profitability, customer satisfaction, and sustainability. The development and evaluation of a digital inventory system for Serai Paradigm outlet not only address immediate operational challenges but also demonstrate the broader significance of adopting digital solutions in ensuring competitiveness and resilience in a rapidly evolving industry.

1.4 Research scope

This study aims to develop and evaluate a digital inventory system for kitchen operations, a solution that directly addresses the needs of the hospitality and food & beverage (F&B) industry. In restaurants, cafés, and hotel kitchens, inventory management is central to ensuring uninterrupted service delivery, cost efficiency, and customer satisfaction. By supplanting manual inventory management techniques, the system seeks to overcome common challenges such as missing data records, human errors in data entry, and delays in acquiring up to date stock information all of which are particularly disruptive in fast paced hospitality environments.

In the F&B industry, the reliance on manual stock monitoring often leads to inconsistencies between recorded and actual stock levels, resulting in food shortages, over purchasing, and unnecessary waste. For example, if kitchen staff record stock inaccurately or fail to update inventory promptly, restaurants risk running out of essential ingredients during peak hours, which directly affects menu availability and customer experience. Conversely, ordering excess stock without real time monitoring leads to spoilage and financial losses. Thus, implementing a digital inventory system

offers a strategic solution by ensuring real time visibility, accuracy, and accessibility of stock information across multiple users and devices.

The scope of this study encompasses the primary investigation into the development and evaluate of systems for real time stock reception, recording, and counting. Within hospitality settings, this means that every stage of kitchen operations from receiving raw materials from suppliers, to storing them, to utilizing them in food preparation can be digitally tracked. Real time updates allow chefs, managers, and procurement staff to make timely decisions on reordering, menu adjustments, and supplier negotiations. Moreover, the digital nature of the system ensures that information is not limited to on site access but can be retrieved from anywhere, a feature that enhances managerial flexibility and oversight.

Although this study involves develop the system in a simulated real time operation, its design is highly relevant to the hospitality industry's long term digital transformation. While many large hotel chains and international franchises already employ sophisticated digital systems, small, medium and also big sized businesses such as individual restaurants and local outlets often lack access to such technologies due to financial and technical barriers. By demonstrating the feasibility and benefits of a digital inventory system in a smaller setting, this study provides a foundation for broader adoption in the future.

In conclusion, this study contributes significantly to the hospitality and F&B industry by addressing critical pain points in kitchen operations through the develop of a digital inventory system. By moving beyond manual processes, the system enhances accuracy, timeliness, and efficiency while laying the groundwork for future scalability and broader application across the sector. Its relevance lies not only in solving operational challenges at the outlet level but also in demonstrating how digital transformation can reshape inventory management practices in hospitality to meet modern industry demands.

1.5 Research significant

This study was conducted to satisfy the researcher's final year project, a requisite for the completion of studies and graduation. Beyond fulfilling academic requirements, the study can potentially provide substantial support to many responsible and interested stakeholders in the hospitality and food & beverage (F&B) industry.

Included among them is the Serai Group company itself. The developed digital inventory system will enhance stock management. It can indeed minimize raw material waste by monitoring current inventory and delivering precise data for sales recording and report preparation. This solution will streamline employees' everyday stock management responsibilities and assist them in ordering products based on real-time inventory levels.

Subsequently, to scholars and learners. This work contributes significantly to the academic domain, particularly within information technology in the food and beverage sector, by introducing practical applications for the advancement of digital inventory systems. Simultaneously, it serves as a resource and reference for students aspiring to undertake system development in either the same or diverse domains.

Ultimately, to prospective researchers. This will create opportunities and act as a framework for the advancement of more complex systems by incorporating technology alongside various elements such as data analytics, artificial intelligence (AI), and integration with e-commerce platforms, thereby facilitating the transformation of both the national and global economy.

1.6 Summary

This chapter succinctly delineates the study design, problem statement, research purpose, research scope, and research significance. These are fundamental elements essential for researchers to acquire clearer information during the course of their study. The conclusion of this chapter is that the conducted study does not provide a practical solution to the researched problem; rather, it offers advantages to users and organizations and acts as a reference for future development and research, particularly in technology and inventory management.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

A literature review provides a comprehensive overview of the sources utilized in the examination of a certain issue. Furthermore, a literature review is an academic composition that elucidates the knowledge and comprehension of the scholarly subject under investigation.

2.2 Research of key concepts and theories

2.2.1 Inventory management

Every organization possesses inventory that must be maintained in storage to facilitate business operations or services. If a corporation neglects inventory management, studies assert that it will have significant challenges in maintaining effective corporate operations and services. Consequently, every organization must implement an effective inventory management system.

Inventory refers to dormant stock that is stored in a warehouse, awaiting utilization (Muhamad & Othman, 2022). Inventory management systems assist organizations in ordering, receiving materials, regulating material flow, and can save costs while optimizing operational efficiency. This method can potentially yield beneficial outcomes by conserving time in inventory management. The additional time gained from optimizing stock management might be allocated to other everyday responsibilities. Moreover, this inventory system enables organizations to observe daily stock usage trends and optimize storage space utilization.

The transformations in the digital age have altered information storage methods, necessitating distinct tactics from those previously employed. The recorded information and data must be effectively handled to ensure quick accessibility, thereby serving as a catalyst for a competitive business. This inventory management system is essential for enhancing high-level information interaction. Hashim and Mokhtar (2021) asserted in their study that organizational failure is attributable not just to external challenges but also to internal issues, including an unsystematic management system, inadequate managerial expertise, and poor management practices.

The delivery and storage of inventory information has evolved throughout time in accordance with the quickly advancing digital age. Prior to the improvements, the majority of information was manually recorded in files or books and thereafter stored in designated locations. This strategy is widely recognized as less efficient in light of recent technology advancements and necessitates revision.

A substantial amount of inventory data and information need storage, which can be accomplished through numerous means and formats. One of them is a digital database (web-based). A web-based inventory management system facilitates organizational operations and is essential for achieving primary objectives, hence enhancing the efficiency and effectiveness of inventory stock storage.

2.2.2 Database-driven systems and web applications

In the current IR4.0 era, internet access can be easily achieved across borders. Information is now at our fingertips, whether for the purposes of education, research, decision-making, performance monitoring, reporting, and also improvements to increase productivity or service efficiency (Aziz, Yusof, Hashim, Rahman, & Mohd, 2024).

Information data management in the contemporary era of globalization has numerous issues due to the advent of big data. The issues include the necessity for extensive data storage, vulnerability to cyber assaults, receipt of data from diverse sources leading to inconsistencies, and variations in legal frameworks and data security measures. Nevertheless, effective software development, coupled with the newest technical advancements, might mitigate these difficulties through the utilization of

cloud-based technologies. While it aids in tackling difficulties, it is ineffectual if the system lacks an efficient and functional database, as this is a crucial component for administering the organization.

This method has the advantage of enhancing individuals' knowledge in integrating databases with advanced and cutting-edge technologies. If the operator comprehends the system's functionality, data management and analysis can expedite the decision-making process. This system additionally provides specialized experts in research and development (R&D) to create systems compatible with new algorithms and data management techniques, capable of generating unique ideas.

Continuing education is vital in the ever-evolving, global world. The capacity to acquire knowledge and adjust to contemporary technologies and data management techniques is crucial. Effective data management and education on data handling and analysis enable individuals and companies to enhance efficiency, lower expenses, and make better informed decisions (Aziz, Yusof, Hashim, Rahman, & Mohd, 2024). Consequently, data management reliant on database and web applications must serve as the foundation and require advanced abilities to compete, create, and thrive in the future.

2.3 Empirical previous research

This section elucidates the studies that have been previously undertaken. This research is conducted by researchers to aid in the development of a product or service. This part will address many aspects to enhance comprehension of the material.

2.3.1 Manual inventory system

A manual inventory system is a form of stock management that does not utilize any technological systems. All information is preserved via conventional techniques, like books, forms, or files. This strategy is straightforward to implement as it does not necessitate the participation of technology specialists and incurs minimal maintenance expenses. Furthermore, for those with limited proficiency in electronic systems, this manual inventory approach remains somewhat independent of advanced technology.

This solution is appropriate for micro or small enterprises. Despite its user-friendliness, there are repercussions that will affect the organization's inventory management. This approach is inefficient for large businesses due to its propensity for errors and prolonged data recording times.

2.3.2 Automated inventory management system

This system represents a contemporary approach aligned with current technical advancements. This technique facilitates inventory stock management, enhancing efficiency and conserving time. Furthermore, it can help reduce errors committed by human resources. This method has features that render it suitable for a corporation, including efficiency, precision, and accessibility.

A study (Ratih Purwasih & Candana, 2024) on the development of an Inventory Management Information System (IMIS) revealed that 85% of respondents concurred that the system was user-friendly and enhanced efficiency in inventory stock management. The mistake rate in maintaining data associated with actual stock dropped from 15% to 3%. Subsequently, regarding operational efficiency, the time required to document data using this inventory system might be diminished by as much as 40%.

2.3.3 Digital technology

Digital technology has catalyzed economic progress, transforming the realm of information without constraints. In recent years, digital technology has garnered worldwide attention due to the advent of innovations such as blockchain, 3D printing, the Internet of Things (IoT), 5G, cloud computing, automation and robotics, and artificial intelligence (AI).

According to data from the China Academy of Information and Communications Technology, the total digital economy of the G20 countries reached \$26.17 trillion in 2017, an increase of 8.64% from 2016, significantly ahead of the Gross Domestic Product (GDP) growth rate over the same period (Wang, Chen, Ramsey, & Hewings, 2021).

2.4 Current practices in the industry

(Ts. Izza Shafini Binti Zainal Abidin, Nurul Izza Binti Mahadan, & Noorhaneza Binti Ariffin, 2025) Mobile technology often pertains to portable and personal devices, such as smartphones. Global technology advancements are continually increasing, enhancing their competitiveness. This technology is accessible at any time and from any location, given an internet connection. A multitude of mobile applications have been created to enhance the daily tasks of users. The swift advancement of the global economy has rendered the development of technology and intelligent applications a primary strategy for enhancing different sectors, including the food and beverage business.

The use of mobile applications greatly helps organizations and individuals access information flexibly and facilitates the process of preparing reports. In addition, it functions as a medium for storing document data related to technical guidelines. Therefore, by developing a digital application or system that integrates data into its system, it has the opportunity to become an important tool in inventory stock control. Therefore, this allows organizations to implement an inventory system in a more organized, systematic way, make decisions based on real stock and can increase its effectiveness in the industry.

A distinct feature of digital economy development is the innovation and replacement of obsolete economic forms by economic entities through digitization with more advanced forms. This procedure is referred to as "digital transformation". Digital transformation in the food and beverage sector entails the application of information and communication technology to enhance operational efficiency, productivity, and service quality (Bambang Suharto, Medina, & Ardianto, 2024). Consequently, the utilization of Point of Sale (POS) systems might be associated with the implementation of digital inventory systems. The integration of the two systems will facilitate improved business operations.

2.5 Knowledge gap

Every organization engaged in diverse operations must own an inventory system. Inventory systems are utilized to manage stock; yet, some firms still employ manual techniques for recording stock data. The majority are small and medium-sized enterprises. Furthermore, the majority of prior research has concentrated on this system inside large-scale enterprises. This aspect is affected by multiple elements, notably that the facilities and technology of major enterprises are more comprehensive than those of small and medium-sized organizations, which possess limited infrastructure. The subsequent consideration is the financial aspect. Organizations must compensate a team of developers to construct a digital inventory system that facilitates inventory management. It is therefore unsurprising that prior studies have predominantly concentrated on major enterprises, as they engage with organizations that are established and stable within the industry. Nonetheless, substantial enterprises continue to employ antiquated methods, specifically manual processes, to minimize operational costs. Small and medium-sized enterprises often prefer manual stock management due to fluctuating and increasing financial constraints.

Consequently, a deficiency in understanding exists regarding the implementation of a digital inventory system to mitigate stock and financial waste. The primary objective of building and utilizing this system is to effectively and systematically address the existing gap.

2.6 Summary

The inventory system is crucial for an organization. Research indicates that it is a systematic and effective framework that facilitates seamless operations and can lower expenses. Previous studies indicate that employing a digital inventory system can reduce human errors, expedite the calculating process, and yield precise data. The established system confidently facilitates data analysis to anticipate demand for products exhibiting significant sales trends.

Consequently, the data from the previous session indicates a substantial gap in the utilization of digital inventory systems. This study is crucial for building a system

that addresses cost efficiency and current requirements, enhances operational efficacy, and enables firms to expand and remain competitive in a continually evolving industry.

CHAPTER 3

METHODOLOGY

3.1 Introduction

The research methodology outlines the techniques and strategies employed in this investigation. The document encompasses the study design, data collection techniques, data analysis methodologies, and anticipated outcomes. Furthermore, the technique is crucial for guaranteeing that the study's aims and outcomes may be attained through the aforementioned variables.

3.2 Research design

Research design is a systematic framework that specifies the details of the study being conducted. It provides a structured guideline that clarifies the methodology, data collection techniques, analytical approaches, and the expected research outcomes. In general, research can be classified into two main categories quantitative and qualitative. Quantitative research emphasizes statistical analysis due to its objectivity and reliance on measurable data, while qualitative research provides in-depth exploration of situations or individuals through observation, interviews, or document analysis (Liyana, 2024).

This study adopts a quantitative descriptive research design. The main purpose of this approach is to obtain measurable data that describe user satisfaction and perceptions of the digital inventory system. A descriptive method is more suitable compared to experimental research because the study does not involve manipulation of variables or controlled testing. Instead, it focuses on capturing actual user responses and summarizing them through statistical analysis. This method provides an accurate representation of the system's usability and effectiveness in its real operational context.

To apply this design, a structured questionnaire was developed as the primary data collection instrument. Respondents were asked to evaluate their level of satisfaction and experience with the newly developed system. The collected responses were then analyzed descriptively to determine whether the digital inventory system achieved its intended objectives. Through this approach, the research is able to provide practical and reliable findings that reflect user acceptance and the system's potential to improve efficiency in the hospitality and food & beverage industry.

3.2.1 Prototyping model

In developing a digital system, a development methodology is essential to guarantee it meets the criteria and operates well as a framework. This study employs the prototype model, a variant of the system development models inside the System Development Life Cycle (SDLC).

This model offers researchers and users a framework to comprehend and execute system development, ensuring optimal functionality aligned with user requirements. The prototype approach is an iterative development process, as the system requires continual enhancement based on user feedback. The prototyping method is advantageous when a user is unable to articulate the specifics of the input, process, and output stages, leading to uncertainty among system developers regarding the algorithm's efficiency. This ambiguity complicates the identification of the operating system and the nature of human-machine interaction to be employed (Nur, Bahar, Muhammad Ghifari Arfananda, & Helina Apriyani, 2021).

The prototype model method comprises two techniques, particularly:

1. Evolutionary Prototype

The prototype is developed continuously so that the functions and procedures produced are following the needs of the system are being developed.

2. Prototype Requirements

Prototypes are made by developers by defining system functions and procedures where users or system owners cannot define system requirements properly.

In the hospitality and food & beverage industry, the prototype model is highly effective. For example, when developing a digital inventory system for a restaurant or hotel kitchen, the first prototype might allow staff to log raw materials and generate simple stock reports. After testing, the chef may suggest adding automatic low stock alerts, while the storekeeper may request supplier information integration. These suggestions are then implemented in the next prototype. The iterative process ensures the final system is practical, user oriented, and aligned with actual kitchen operations.

3.2.2 Prototyping model workflow

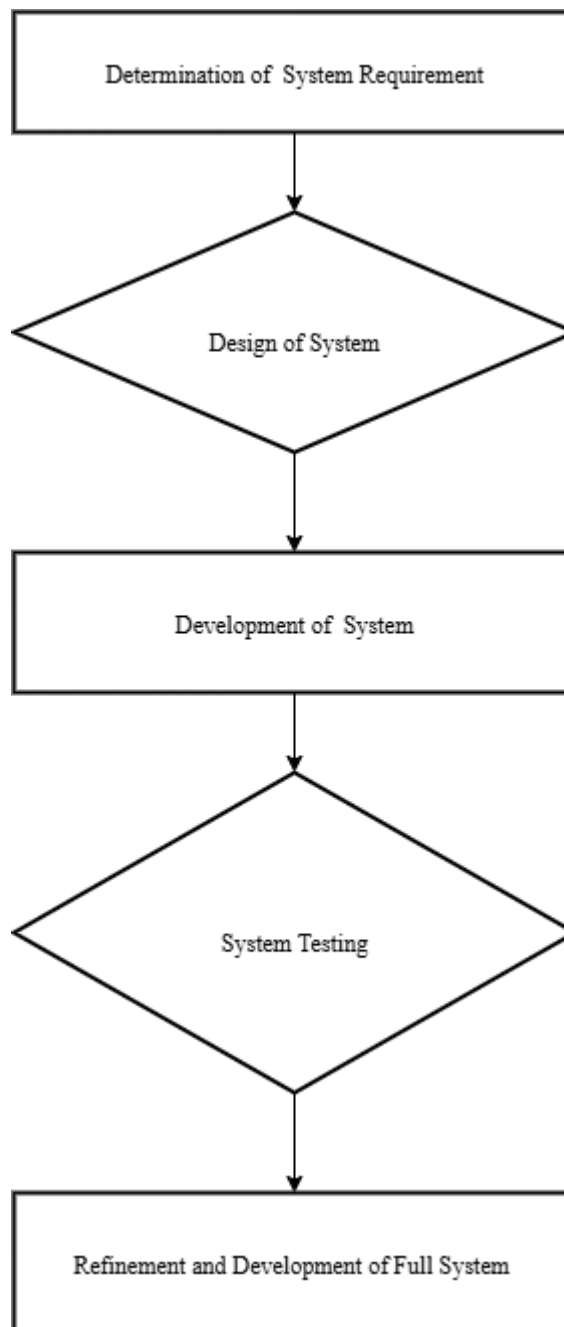


Figure 3.1 Prototyping model

The prototype model demonstrates an iterative approach to developing a digital inventory system. In the first phase, determination of system requirement, the researcher gathered information on kitchen operations at Serai Paradigm Outlet, focusing on challenges in tracking stock, reducing waste, and minimizing human error.

Next, a design of the digital system was developed, which included basic features such as raw material input forms and automatic stock updates. This prototype was then subjected to system testing, where staff such as chefs and storekeepers tested the system and provided feedback on usability and functionality. Based on their input, the system underwent refinement, with improvements made to reporting features, interface design, and accuracy of stock tracking. Finally, the process concluded with the development of the full system, a reliable digital inventory solution designed to enhance efficiency, reduce wastage, and support decision making in hospitality and F&B operations.

3.3 Data collection

Data collecting methods are procedures employed to acquire the necessary data and information to address research questions and fulfill research objectives. This technique is crucial to ascertain the study's outcomes, determining its success or failure. Consequently, data gathering procedures significantly impact respondents' trust and the effectiveness of the research conducted.

The questionnaire was employed as one of the main data collection methods in this study, as it is a widely recognized tool in quantitative research for systematically gathering information from respondents through structured questions. Its primary purpose was to obtain direct feedback from participants regarding their experiences, perceptions, and practices in relation to the research topic. This method served as a complement to observation and document analysis, allowing the researcher to cross-validate findings and strengthen the reliability of the study. Questionnaires are generally available in two formats, namely paper-based and digital, such as those distributed through online platforms like Google Forms. For this research, a paper-based categorical questionnaire was selected to avoid disruptions caused by internet connectivity issues or reliance on email accounts, ensuring greater accessibility and participation.

The design of the questionnaire emphasized categorical questions, which encouraged respondents to provide specific and standardized answers. This format simplified the data analysis process by enabling responses to be quantified and

compared easily, while also reducing the risk of ambiguity or misinterpretation. Furthermore, the use of a paper-based questionnaire was particularly practical in the hospitality and food and beverage environment, as it allowed for straightforward distribution among staff and stakeholders. In addition, the standardized nature of the questions promoted consistency across responses, enhancing the validity of the results. Overall, the questionnaire method played an essential role in ensuring the collection of reliable and structured data that contributed significantly to achieving the objectives of the study.

3.3.1 Data collection flow

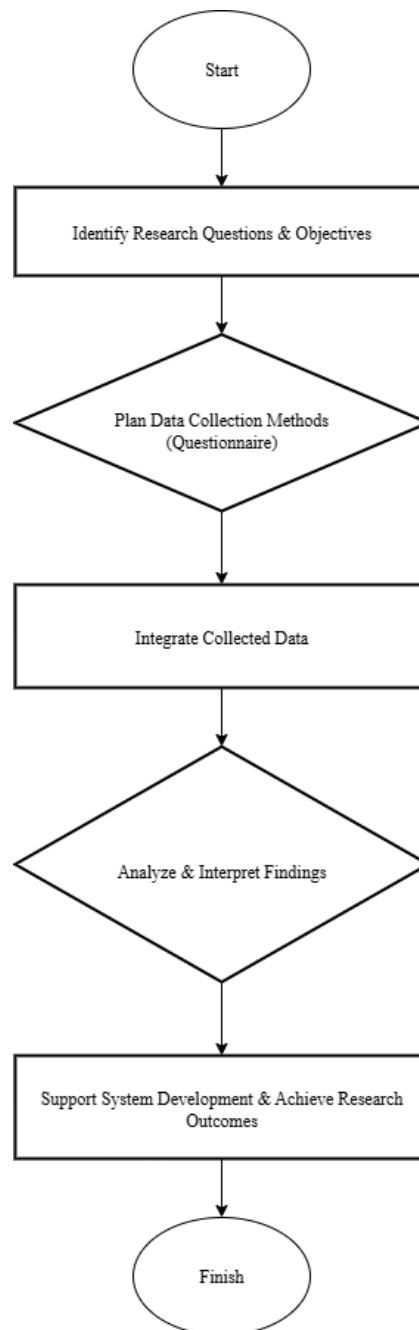


Figure 3.2 Data collection flow

The flowchart shows how the data gathering process in this study was organized. The first step is to figure out what the research questions and goals are, which will guide the whole process. After they are set up, the next step is to figure out the best ways to gather data, with the questionnaire being a major instrument. At this point, the

researcher carefully crafts the questionnaire so that it can get useful answers from participants. This planning procedure makes sure that the instrument meets the goals and fully answers the research questions.

After the planning stage, the data that was gathered is put together and structured in a way that makes it ready for more analysis. During the analysis and interpretation step, the researcher might look for patterns, make conclusions, and judge how important the results are. In the end, the data that was evaluated supports the creation of a computerized system and helps the research goals be met. The workflow shows how each phase is connected and necessary for getting useful and accurate research results by moving from problem identification to data gathering, integration, analysis, and finally system implementation.

3.4 Analysis data

Data collection must be conducted in phases to ensure proper and orderly acquisition of information. Upon completion of all techniques, the acquired data will be transposed into Microsoft Word as a pie chart to elucidate the results. The pie chart is employed to assess the feedback from the respondents' sample.

The total number of participants is 15 individuals. The researcher will select 14 responders from that total as a sample to get input pertinent to the study undertaken. The sample size established is derived from the Krejcie and Morgan table. The study sample comprises individuals and locations that provide the requisite information for the researcher. The sample size must be less than the whole population to facilitate data analysis by the researcher.

3.4.1 Sample size determination

Krejcie and Morgan (1970) proposed a formula for determining sample size when the population is known. The formula is written as:

$$S = \frac{\chi^2 \cdot N \cdot P(1 - P)}{d^2(N - 1) + \chi^2 \cdot P(1 - P)}$$

In words, the formula states that the sample size (S) is calculated by multiplying the chi-square value (χ^2) with the population size (N) and the proportion of the population expected to choose one option (P) times its complement (1-P). This value is then divided by the sum of the margin of error squared (d^2) times the population size minus one (N-1), added to the chi-square value (χ^2) multiplied again with P(1-P). In this formula:

- S = required sample size
- N = population size
- P = population proportion, usually assumed as 0.50 to maximize sample size
- d = degree of accuracy or margin of error, typically set at 0.05
- χ^2 = chi-square value for 1 degree of freedom at the desired confidence level, commonly 3.841 for 95% confidence

Based on Krejcie and Morgan's (1970) formula for determining sample size, the calculation was conducted using a population size of fifteen (N=15). With a 95% confidence level ($\chi^2 = 3.841$), a margin of error of 0.05 ($d = 0.05$), and assuming the maximum population proportion ($P = 0.50$), the resulting computation produced a sample size of approximately 14.48. When rounded to the nearest whole number, the required sample size is 14.

3.5 Expected outcome

The researcher anticipates a beneficial effect on the company and its services based on the study. One anticipated impact is enhanced operating efficiency. This will expedite the processes of recording, updating, and monitoring inventory relative to the previous way. Furthermore, it can mitigate human errors occurring during data recording. Errors in inventory estimations and missing data are anticipated to be reduced. Subsequently, it has the potential to conserve time and reduce expenses. The time allocated for stock

management can be reduced, allowing for the reallocation of resources to other tasks and ultimately decreasing management expenses over time. Moreover, it will facilitate more accessible data retrieval. This is due to its enhanced accessibility and precision for management and users in making inventory order selections. This study regularly benefits corporations for future applications. Additionally, it offers pragmatic recommendations for firms to manage inventories in a more methodical and effective manner.

In conclusion, it is anticipated that this study will be approved to assure the achievement and broad application of its objectives and goals.

CHAPTER 4

PLANNING

4.1 Gantt chart

Bil	Task/Planning	Month / Week											
		Jul 2025				Aug 2025				Sept 2025			
		1	2	3	4	1	2	3	4	1	2	3	4
1.	Introduction to Final Year Project 1	■	■										
2.	Planning a proposal paper		■	■									
3.	Abstract writing			■	■								
4.	Writing chapter 1 Introduction				■	■							
5.	Writing chapter 2 Literature review					■	■						
6.	Writing chapter 3 Methodology							■	■				
7.	Presentation proposal paper								■	■			
8.	Correction of final report										■	■	
9.	Submission of final report											■	■
10.	Final year project 1 finish												■

Table 4.1 Gantt Chart

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APPENDICES

11/08/20

DAILY KITCHEN AM & PM INVENTORY

DATE	ITEM	QUANTITY	RECEIVED STOCK	ISSUED STOCK	CLOSING STOCK	TOTAL	REMARKS
11/08/20	AYAM						
	AYAM	PCS	119	60	129	46	
	PURFO	PCS	5	-	45	5	
	LAMB SHANE	PCS	12	-	9	2	
	RIB EYE	PCS	1	22	22	1	
	SALMON	PCS	38	5	41	1	
	TILOPIA	PCS	8 1/2	-	1	7	
	PRONUT	PCS	2 1/2	-	4 1/2		
	vdang GARDIN	PCS	26	11	34	2	
	SIAKAP M	NOS	11	#	4		
SIAKAP L	NOS	5		5			
GAROUA	NOS	2		2			
12/08/20							

MRS. SIAKAP

Daily inventory form